

Field Report: McGee Creek Surficial Geomorphology and Active Tectonics

Abstract

The surficial geomorphology of McGee Creek in the Eastern Sierras preserves a history of extensive intrusive volcanism, glaciation and faulting. Well-preserved moraines of unconsolidated glacial till associated with the Tioga glaciation during the last glacial maximum (MIS 2) and the Tahoe glaciation, of MIS 4-6, are bisected by the Hilton Creek Fault, a major range front normal fault. Significant oblique right lateral faulting within the field area has produced numerous fault scarps from which we can quantify fault offsets, slip rates and their direction. Our study quantified slip along the Hilton Creek Fault to be 1.77 cm/yr with a total slip displacement of 22.5 m, in addition to reinforcing the younger age of the Tioga glaciation relative to the Tahoe glaciation using structural, erosional and crosscutting relationships.

Introduction

McGee Creek is a U-shaped glacier carved valley that sits just east of the Mammoth Lakes and just south of the Benton Range. The Hilton Creek Fault, of Walker Lane and the associated Sierra Nevada frontal fault system, is exposed at McGee Creek and is a byproduct of dextral motion between the Pacific plate and the Basin and Range province extension in the North American Plate. Steep normal faulting and strike slip shearing are the most common structures within the fault system, and are hypothesized to accommodate microscopic microplate translation westward, in lieu of Sierran Uplift or Basin Range extension (Unruh, 2003). The Hilton Creek Fault forms steep fault scarps where it crosscuts moraines and surficial deposits associated with the Tioga-age (Marine Isotope Stage 2) last glacial maximum (LGM, 27-15 ka) and Penultimate Tahoe-age (Marine Isotope Stage 4-6) glaciation (145-130 ka), suggesting a Holocene age for the fault system (Rood et al., 2011, Berry, 1997, and Blackwelder, 1931). Mapping the faulted moraines provides an opportunity to constrain the structural relationships between the fault and the moraines, specifically the amount of fault offset, the direction and rate of motion, as well as the temporal relationship between moraines.

Geologic Background

Cretaceous intrusion of Round Valley Peak granodiorite plutons into Ordovician sediments produced massive hills capped with metasedimentary units of hornfels facies marbles, shales, and slates from contact metamorphism. Accumulation and subsequent ablation of Tahoe and Tioga-aged glaciers severely eroded these igneous and metamorphic bodies and produced massive glacial moraines deposits of glacial till, while hillslope erosion and melt processes formed colluvium and alluvium, respectively.

Hilton Creek Fault

The 22 km Hilton Creek Fault is one of two major Sierra Nevada range-front faults that bound Long Valley Caldera. The fault is directed N/S with an azimuth of $\sim 60^\circ$ and a dip ranging from 30° to 70° (Fig. 1, Berry, 1997). The fault scarps indicate increased vertical slip northward, though the size of individual displacements are unconstrained. Berry (1997) proposed 7-9 faulting events with 3-4 m per event (typical for normal faults) displacement likely occurred over the last 25-20 ka, with a major surface faulting event occurring within the last 38 ka. The average fault slip rate over the last 16 ka is calculated to be ~ 1.1 mm/yr (Hildreth et al., 2018). Hilton Creek is characterized as an oblique normal fault with a right lateral component to the northeast (Unruh et al. 2003, Berry 1997).

Tahoe Glaciation

The oldest moraines within McGee Creek are aged to be $\sim 144 \pm 14$ ka via ^{10}Be and ^{36}Cl cosmogenic nuclide exposure dating and are associated with the Tahoe-aged (MIS 4-6) glaciation (Rood et al. 2011). These moraines contain significantly weathered and fractured metasedimentary hornfels, quartzite, slate, and marble in addition to igneous clasts of granodiorite with diorite xenoliths derived from the Round Valley Peak and the slopes of the mountains bounding the valley (Hildreth et al., 2018). Analysis of Tahoe-aged deposits suggest slip rates almost twice as high as those of the modern Hilton Creek Fault. Fault scarps within the mapped area are frequently on poorly sorted deposits of boulders in conjunction with soils high in pedogenic carbonate that can maintain a steep angle of repose for 'long periods of time' (Berry, 1997).

Tioga Glaciation

The youngest moraines within the valley are associated with Tioga-aged glaciation and have been dated at 18.8 ± 1.9 ka via cosmogenic nuclide exposure dating (Rood et al. 2011). The clasts of the Tioga moraines display an almost identical source and composition of granodiorite and hornfels facies metamorphism to those of the older Tahoe moraine, except they are significantly less weathered and on average larger than their counterparts. Wahrhaftig et al., 2019 theorized the maximum height of the Tioga icesheet to be greater than 2700 m.

Methods

Fault and bedding plane measurements and surface deposit mapping was conducted using Fieldmove Clino on an iPad, from which inferred unit extents were determined. Clast count observations of lithology, weathering and size were obtained using a hand lens, meter stick and dilute HCL acid. Fault scarp slope survey measurements were produced using a sighting level, Brunton compass, metering tape and a stadial rod. A LiDAR extracted digital elevation model (DEM) was used for comparison to field slope observations, to ensure accuracy of in-field measurements. Python 3 and QGIS were used for data analysis of dip-slip distance, lateral

offsets, and total slip along the Hilton Creek Fault plane as well as for comparative analysis of clasts within the Tioga (younger) and Tahoe (older) moraines. Localized erosion around boulders and brush cover have produced variable slope angles which requires maximum paleo-slope angles and displacements to be calculated mathematically in conjunction with field observations using techniques from Berry (1997).

Results and Discussion

A geological map showing the inferred extent of glacial and fluvial surficial deposits associated with the Tioga and Tahoe glaciations as well as bedrock geology exposed within the McGee Creek study area was produced to better understand the structural relationships present in the field area (Fig. 1A). The fault is directed N/S with an azimuth of $\sim 60^\circ$ and a dip ranging from 30° to 70° dependent on rheology. Faulting within the study area was observed to be normal, with an oblique right lateral component supported by linear orientations taken from slickensides on the granodiorite and metasediments that displayed an average trend of 66° with a plunge of 40° . The A-A' topographic profile (Fig. 2A) shows that the crests of the younger Tioga moraines are much steeper, $\pm 40^\circ$, than their Tahoe counterparts, with slope angles of $\pm 20^\circ$. 200 m slope surveys conducted across the fault scarp both by hand and using LiDAR imaging also found the scarp to have an average slope angle of $\pm 40^\circ$ (Fig. 2B). Our measured fault offset along the younger Tioga moraine had a vertical offset of 9.6 m and an associated dip slip distance along scarp plane of 42.7 m with a dip angle of 25.2° , correlating to a slope of .471 (Fig. 3A). Normal faults typically range from 30° to 60° , so the low slope angle of the scarp is likely due to erosion over the last ~ 19 ka since the LGM. Qprof DEM plane intersection plugin for QGIS allowed for an estimation of the plane associated with Hilton Creek Fault. The measured fault plane produced a dip direction of 68.48° and a dip of 47.94° which correlates with a slope of 1.10 (Fig. 3A). Using the linear regression fit for the paleo-slope of 47.94° combined with the change in vertical elevation along the scarp a dip slip distance of 22.5 m was calculated. A lateral offset of ~ 25 m was measured in QGIS using LiDAR imagery, which in conjunction with the calculated dip slip distance produces a total paleo-slip distance of 33.66 m to the northeast (Fig. 3B). Total slip is ~ 1.77 mm per year for the last ~ 19 ka. Our data agrees well with previous studies of determination of the fault as right-lateral oblique-slip fault with a dip ranging from 30° to 70° (Betty, 1997).

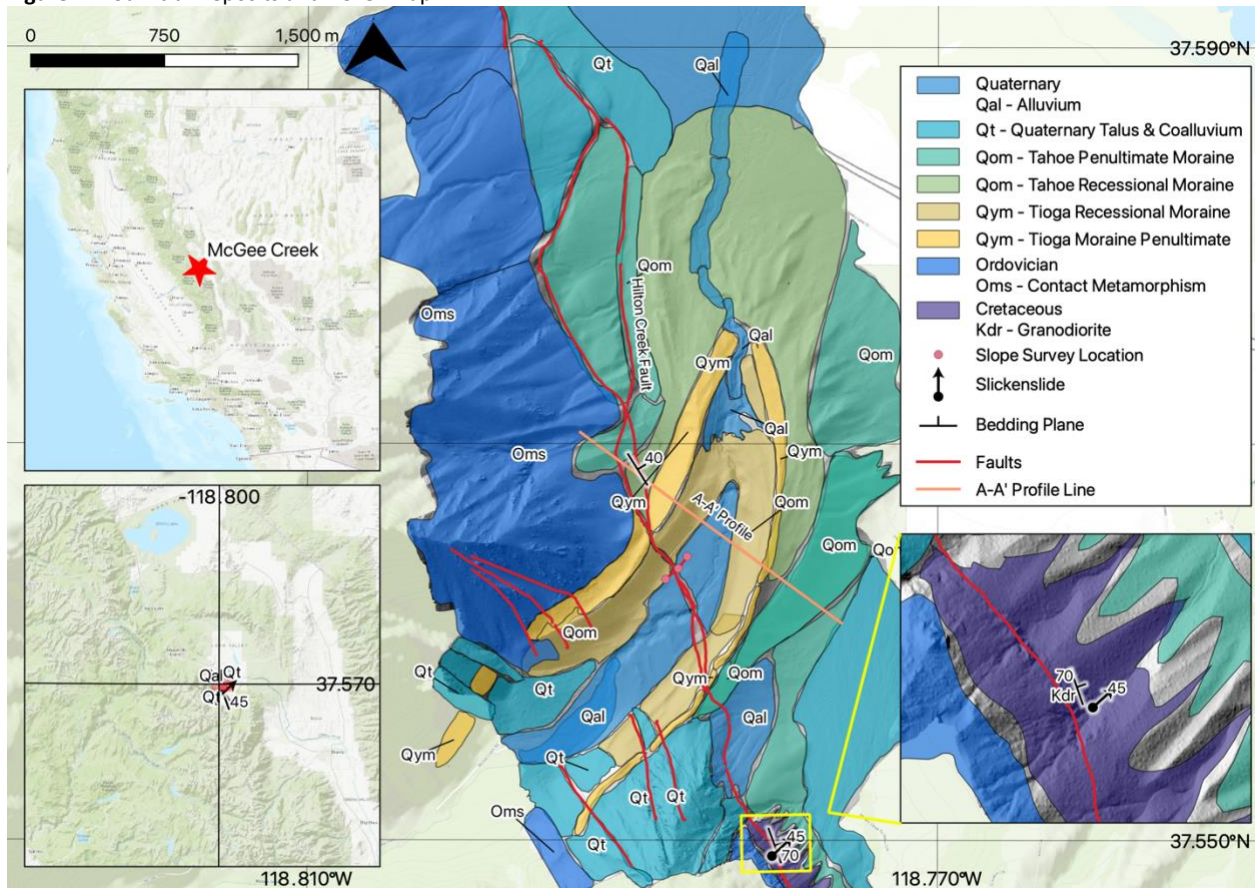
Clast size, amount of weathering and oxidation in conjunction with diorite xenolith protrusion depth within the granodiorites can elucidate the temporal relationships between moraines in addition to cosmogenic nuclide exposure dating. Clasts prominent in the Tahoe moraines were heavily weathered and oxidized, particularly within the metasedimentary units where the long axis of clasts are frequently < 50 cm with most clasts displaying $x < 2.5$ cm. Igneous clasts within the Tahoe moraine have a maximum x axis of ~ 225 cm with most clasts averaging < 30 cm (Fig. 4A and 4B). Fig. 4C reinforces the significant erosional difference between the units, with the younger Tioga moraine displaying clasts with significantly larger area, ≤ 250 cm², than the Tahoe moraines, whose clasts average ~ 6.25 cm², with the largest clasts being ≤ 175 cm². The exposure depth of diorite xenolith protrusions in granodiorite also supports the temporal relationships between the older and younger moraines. Within granodiorites of the younger

Tioga moraine there is a higher density of <5 cm protrusions as opposed to the older moraine which shows a higher density of protrusions with depths >5 cm (Fig. 4D). Measurements of the exposed z-axis of boulders also strengthen the age relationship between moraines. Within younger moraines a z-axis of ~8.5 cm is more often exposed at higher densities than the ~5 cm frequently exposed for boulders within older moraines (Fig. 4E), consistent with a greater amount of burial and sedimentation for older Tahoe-aged moraines. From the mapped polygons in Fig. 1A the well-preserved crests of the Tioga moraines can be observed crosscutting the eroded Tahoe moraines (144 ± 14 ka) reinforcing the younger age of the Tioga glaciation (18.8 ± 14 ka), as supported by previous cosmogenic nuclide dating by Rood et al. (2011).

Conclusion

Structural, erosional and crosscutting relationships between the units all support the Tioga glaciation and its moraines to be younger, evidenced by their low weathering and larger clast area, than the adjacent Tahoe moraines, even exclusive of cosmogenic nuclide dating. The rate of slip through the glacial deposits along the Hilton Creek Fault is quantified at ~1.77 mm/yr. Slow-creep of the fault maybe able to diffuse the dextral stresses over millions of years, but the unconsolidated nature of glacial deposits may produce rheological weaknesses along the fault plane potentially producing an earthquake of M6.8.

Figure 1A: Surficial Deposits and Relief Map



Caption: Inferred extended of surficial geomorphological units exposed within the mapped field area at McGee Creek based on field observations. Over the past ~160 kyr glaciers have eroded Cretaceous granodiorite and Ordovician metasedimentary units to produce significant moraines, in addition to talus and alluvial deposits. The younger Tioga moraine of the LGM (18.8 ± 1.9 ka, yellows) is bound by the older Penultimate Tahoe moraine (144 ± 14 ka, greens). Ages are based on ^{10}Be and ^{36}Cl exposure dating of cosmogenic nuclides trapped in bedrock from spallation reactions with cosmic radiation (Rood et al., 2011).

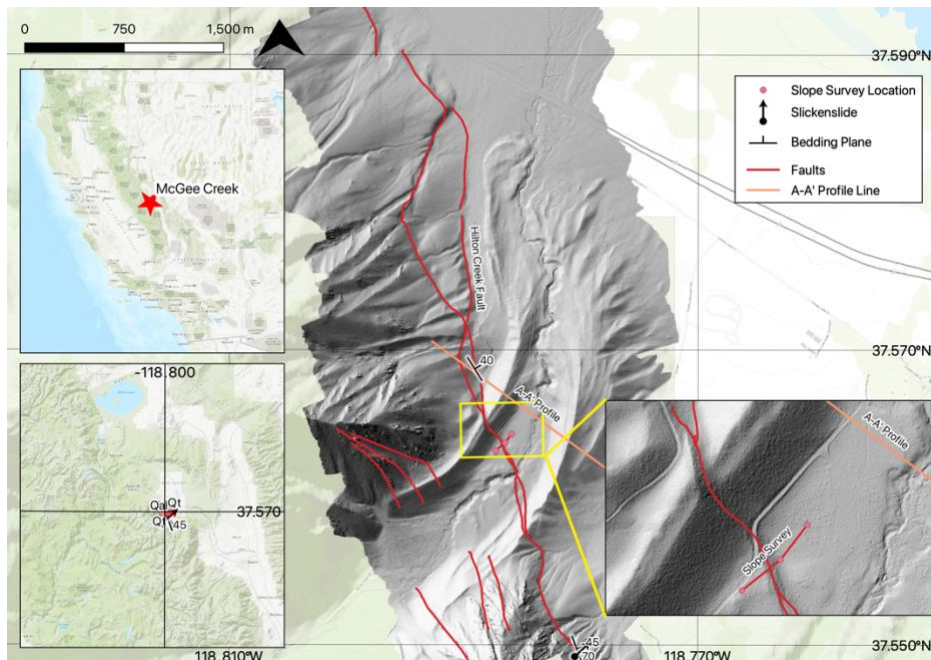
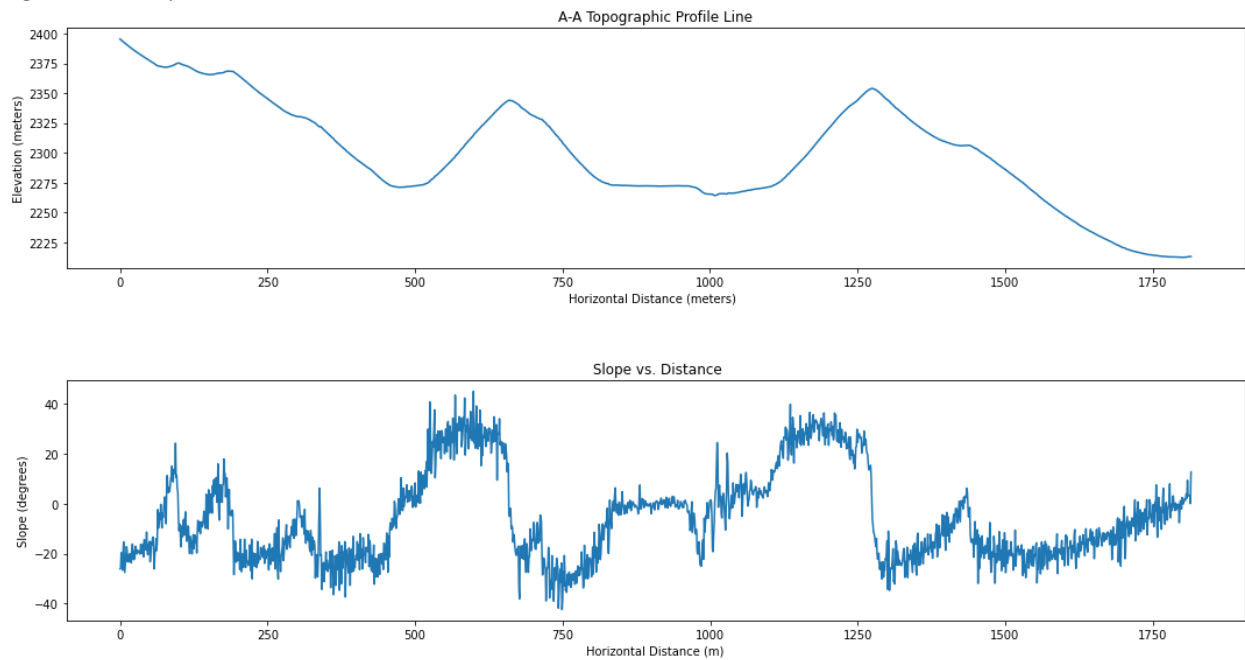


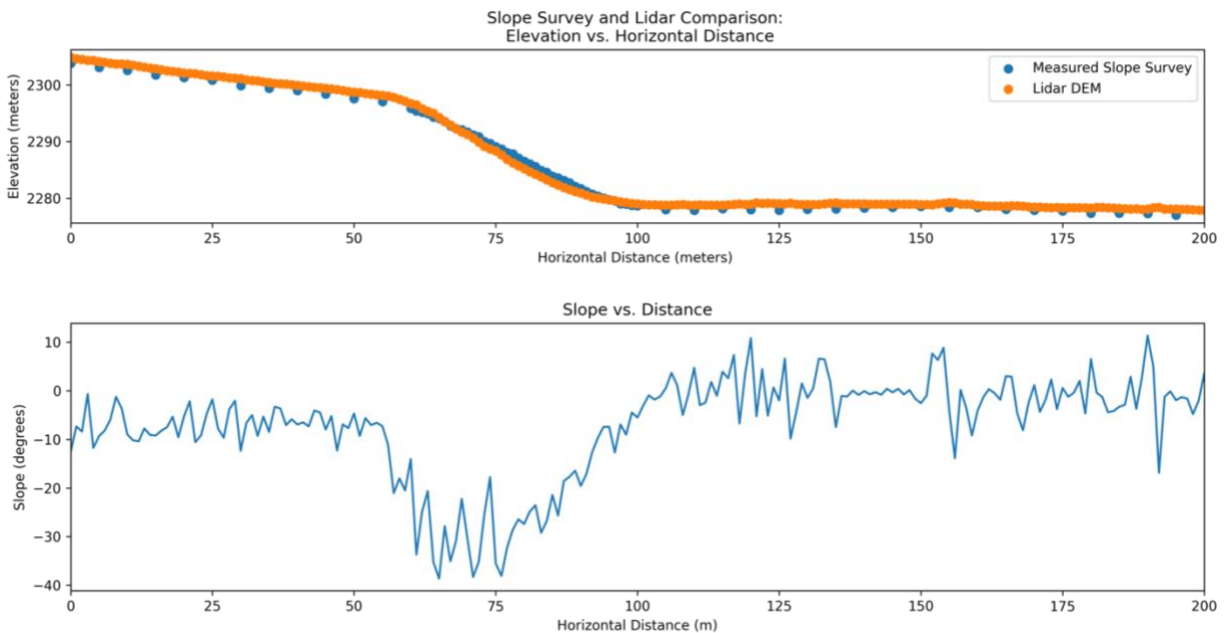
Figure 2B: Surficial relief map that highlights faults within the mapped field area, the A-A' topographic cross section line and our measured hillslope profile along the scarp of the Hilton Creek Fault.

Figure 2A: Hillslope Profiles



Caption: Top: Topographic profile from A-A' (Figure). The small, rounded hills at $x = 275$ m and 1450 m are associated with the Penultimate Tahoe Moraine (age) while the large angular crests at $x = 700$ m and 1300 m belong to the younger Tioga moraine (age). Bottom: Slope angle varies between 0° and $\pm 40^\circ$ across the A-A' profile line, with the steepest slopes associated with the younger Tioga moraine.

Figure 2B: Survey Slope Profiles



Caption: Top: A topographic profile comparison between a hand conducted slope survey over 200 m (this study) and a LiDAR produced slope profile over the same distance, across a fault scarp. Bottom: Both profiles display an almost identical trend in slope change across the same horizontal distance. Slope values range from a minimum of 0° to a maximum of $\sim 40^\circ$ along the fault scarp, consistent with normal faulting.

Figure 3A: Fault Offset and Total Slip

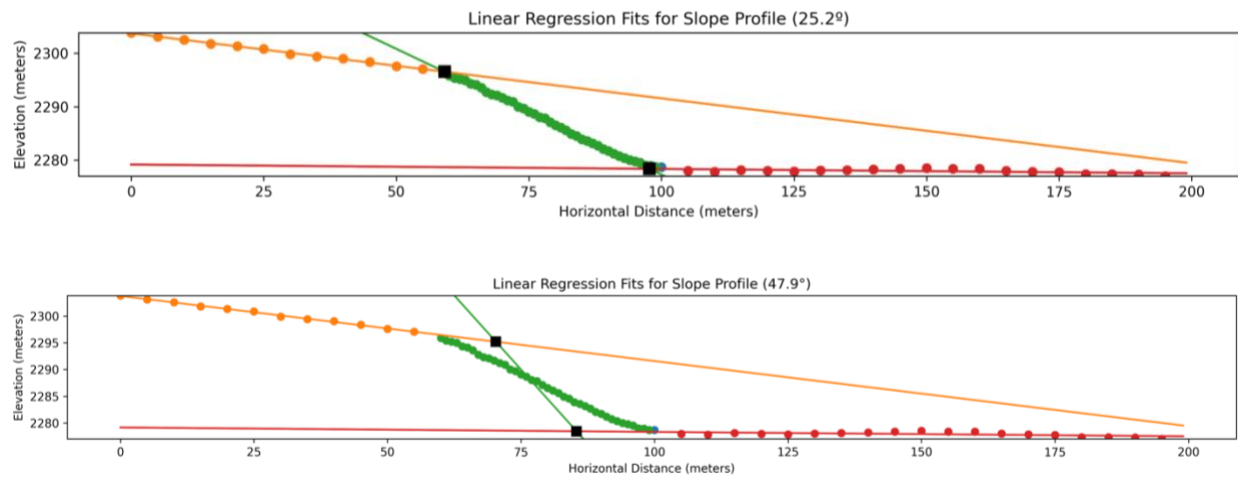
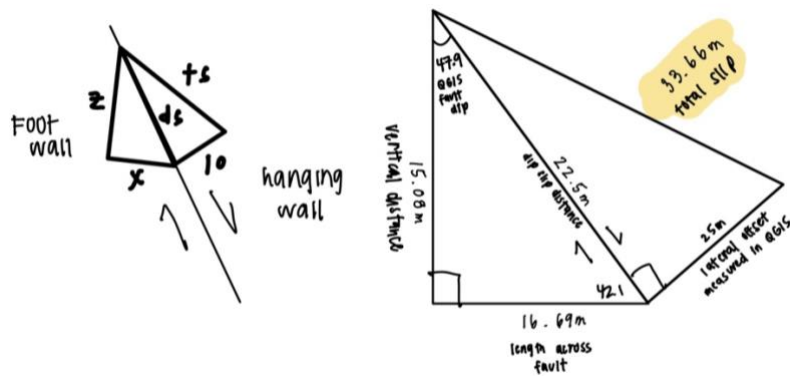


Figure 3B: Dip Slip and Total Slip Calculation



Caption: Top: The measured fault along the younger Tioga moraine has a vertical offset of 9.6 m and the dip slip distance along scarp plane is 42.7 m with a dip angle of 25.2° , correlating to a slope of .471. The low slope of the scarp is due to erosion over the last ~19 ka. **Middle:** Qprof DEM plane intersection plugin for QGIS allowed for an estimation of the plane associated with Hilton Creek Fault. The measured fault plane produced a dip direction of 68.48° and a dip of 47.94° which correlates with a slope of 1.10. The dip slip distance of the steeper slope angle is 22.54 m with a lateral offset of 25 m (measured in QGIS), resulting in a total slip distance of 33.66 m NW. Total slip is 1.77 mm per year over ~19 ka. **Bottom:** Using the linear regression fit for the paleo-slope of 47.94° combined with the change in vertical elevation along the scarp a dip slip distance of 22.5 m was calculated. A lateral offset of ~25 m was measured in QGIS using LiDAR imagery, which in conjunction with the calculated dip slip distance produces a total paleo-slip distance of 33.66 m to the northeast.

Figure 4A,B:

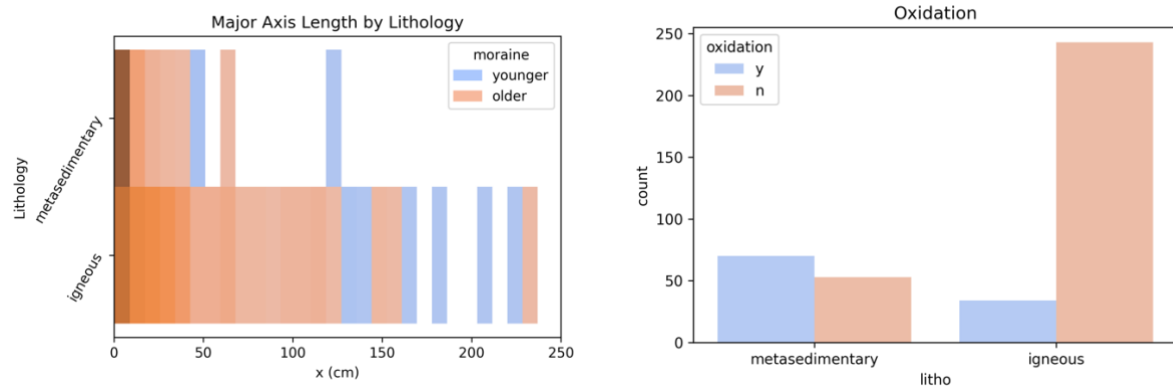


Figure 4C,D:

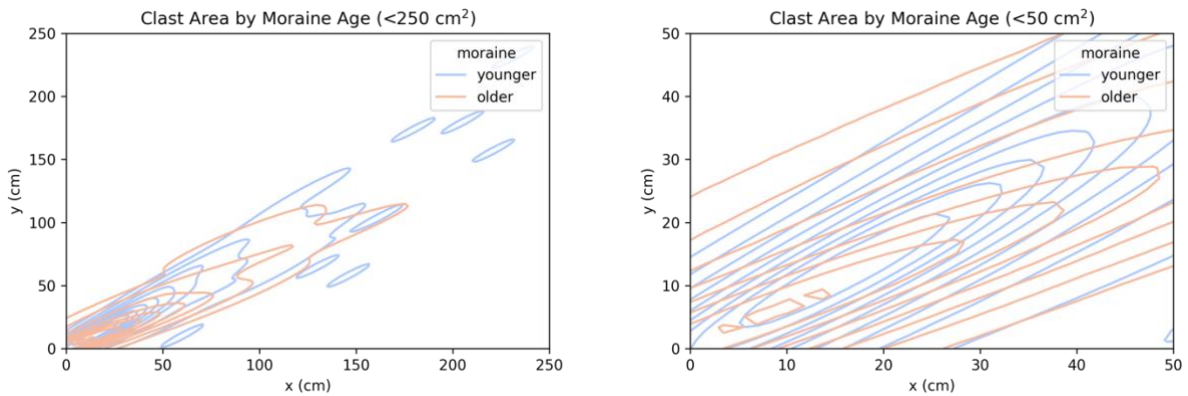
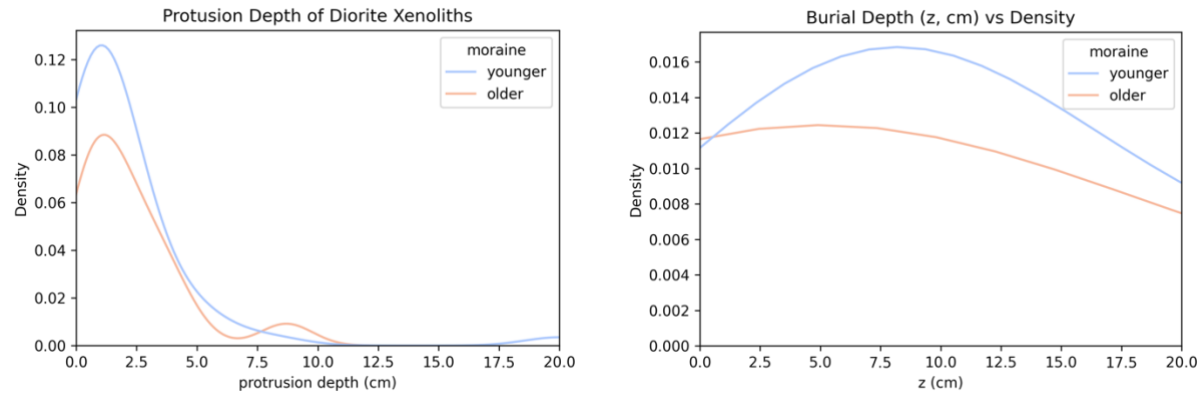


Figure 4D,E:



Caption: **A:** Figure shows a histogram of the length of the major axis of clasts based on lithology sorted by moraine age. The younger moraine clasts consistently display a larger size than the older moraine. **B:** Histogram shows how oxidation is more prevalent in metasedimentary units which when combined with Fig. 4A shows significant weathering of older metasedimentary units. **C:** A KDE plot comparing the area of clasts from both younger and older moraines. Clasts within the younger moraine consistently display a larger area, which Tahoe clasts comprise the smallest of all clasts. **D:** The figure shows a plot comparing the mean protrusion depths of diorite xenoliths within granodiorites and their density of occurrence. Younger moraine shows a higher density of protrusions with small z-axis relative to Tahoe-aged protrusions. **E:** Figure shows the density of z-axis depth measurements for clasts within the moraines. The younger moraine has a higher density of clasts with a longer z-axis implying they experienced less burial than the older moraine.

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